

NURSING AND MIDWIFERY TRAINING COLLEGE, ATIBIE-KWAHU
REGISTERED MIDWIFERY
MID SEMESTER EXAMS

CLASS: RM 17

DATE: 11/11/2024

TIME ALLOWED: 40 MINUTES

INSTRUCTIONS: Write your index number on all sheets. From question 1 to 40 Circle from letter A to D the appropriate relevant response. (1 mark each)

INDEX:

NAME:

1. A nurse, Emily, notices that accessing a patient's electronic health record (EHR) allows her to see all recent tests, physician notes, and medications prescribed from multiple departments. This integrated view helps her provide more informed care. What core benefit of health informatics does this scenario highlight?
☒ A. Comprehensive access to patient information across departments
☐ B. Enhanced data privacy and security
☐ C. Improved patient engagement
☐ D. Streamlined patient communication
2. Dr. Lee recalls that in the early 1960s, hospitals began experimenting with computers to manage patient information, replacing traditional paper records. Which of the following best describes this historical shift?
☒ A. The beginning of telemedicine
☐ B. The start of electronic health records (EHR) development
☐ C. The invention of mobile health apps
☐ D. The standardization of medical coding
3. In her early days, Florence Nightingale documented patient outcomes and identified patterns that helped improve sanitation practices in hospitals. What early practice in nursing informatics does this scenario demonstrate?
☒ A. Collection and analysis of data to improve patient care
☐ B. Development of nursing theories
☐ C. Establishment of hospital policy management
☐ D. Use of EHRs to streamline patient discharge
4. The ANA defines nursing as involving the "protection, promotion, and optimization of health." Which of the following scenarios best illustrates this component of nursing?
☒ A. A nurse designs hospital administration policies.
☐ B. A nurse educates a community on lifestyle changes to prevent heart disease.
☐ C. A nurse ensures a patient receives their medications on time.
☐ D. A nurse focuses solely on treating acute illnesses.
5. Which component of the ANA definition of nursing emphasizes the role of the nurse as an advocate?
☒ A. Ensuring that hospital policies are followed strictly
☐ B. Focusing on administrative tasks within a healthcare setting
☐ C. Monitoring patient progress without interacting with family members
☐ D. Promoting patient rights and ensuring informed consent
6. How does nursing informatics benefit patient safety?
☒ A. By allowing only senior management to access patient records
☐ B. By limiting patient information access to a single provider
☐ C. By reducing errors through electronic health records (EHRs) and clinical alerts
☐ D. By replacing nursing assessments with automatic data entry

7. Which of the following is a major challenge for nurses and midwives who are new to informatics technology?

- ☐ A. Excessive familiarity with technology
- ☒ B. Lack of adequate training and support in using informatics tools
- ☐ C. Limited opportunities for hands-on patient care
- ☐ D. Overuse of manual record-keeping methods

8. Which of these is a common technical challenge when implementing new informatics systems in nursing?

- ☒ A. Compatibility issues with existing technology and software
- ☐ B. Excessive familiarity with advanced technology
- ☐ C. High satisfaction with existing manual methods
- ☐ D. Limited use of telecommunication in patient care

9. A significant challenge in nursing informatics is interoperability. What does this term refer to?

- ☐ A. Limited access to patient information across different healthcare facilities
- ☐ B. Reduced effectiveness of nursing interventions
- ☒ C. The ability of different systems and technologies to communicate and share data effectively
- ☐ D. The inability to track patient medications across departments

10. Nursing informatics has improved healthcare efficiency by:

- ☐ A. Eliminating all nursing documentation tasks
- ☐ B. Making paper-based records the primary data source
- ☐ C. Reducing the need for clinical assessments
- ☒ D. Streamlining workflows with electronic health records and automated documentation

11. How does nursing informatics improve patient engagement and empowerment?

- ☐ A. By focusing only on clinical data management
- ☐ B. By limiting patient access to healthcare information
- ☒ C. By providing patients with access to their health information and education resources
- ☐ D. By requiring patients to contact healthcare providers for all information

12. In what way does nursing informatics support better allocation of healthcare resources?

- ☒ A. By analyzing data to identify trends and optimize resource distribution
- ☐ B. By eliminating the need for patient records
- ☐ C. By focusing only on administrative resources
- ☐ D. By requiring additional manual data entry

13. The use of predictive analytics in nursing informatics has helped the healthcare system by:

- ☒ A. Identifying at-risk patients for early intervention and improving outcomes
- ☐ B. Limiting treatment options based on historical data
- ☐ C. Reducing the collection of patient data
- ☐ D. Restricting patient information access to doctors only

14. Which aspect of the nursing informatics framework helps support evidence-based practice?

- ☐ A. Theoretical nursing models that exclude technology
- ☒ B. The integration of knowledge management systems
- ☐ C. The limitation of data use in clinical settings
- ☐ D. Replacement of all patient records with manual documentation

23. Operational data in healthcare includes information about:

- ☒ A. Billing and coding information
- ☐ B. Hospital staffing, equipment usage, and workflow
- ☐ C. Insurance claim status
- ☐ D. Patient's clinical diagnoses

24. Which of the following is true about unstructured healthcare data?

- ☒ A. It includes data like physician's handwritten notes or medical images.
- ☐ B. It is easily entered into and analyzed by computerized systems.
- ☐ C. It is typically stored in a numerical format for easy analysis.
- ☐ D. It refers to standardized, coded information.

25. Health insurance claims and coding information fall under which category of healthcare data?

- ☒ A. Administrative data
- ☐ B. Clinical data
- ☐ C. Financial data
- ☐ D. Operational data

26. Which type of data is used to assess the efficiency and effectiveness of healthcare delivery systems?

- ☒ A. Administrative data
- ☐ B. Clinical data
- ☐ C. Financial data
- ☐ D. Operational data

27. How does nursing informatics improve administrative functions in healthcare?

- ☒ A. By automating processes such as scheduling, billing, and resource management to improve efficiency
- ☐ B. By focusing solely on clinical outcomes without considering operational needs
- ☐ C. By limiting communication between the administrative and clinical teams
- ☐ D. By reducing the use of technology to manage patient data

28. In the 1980s, which key development in nursing informatics helped standardize nursing data collection?

- ☐ A. The creation of the first nursing informatics certification program
- ☐ B. The development of the International Classification of Nursing Practice (ICNP)
- ☐ C. The implementation of nursing care plans in electronic format
- ☒ D. The launch of the first computerized nursing database

30. In the 2000s, which technological advancement revolutionized the accessibility and sharing of health information?

- ☐ A. The creation of paper-based hospital charts
- ☐ B. The introduction of fax machines in healthcare
- ☒ C. The invention of the first telemedicine devices
- ☐ D. The widespread use of the internet and online health platforms

31. How does telemedicine enhance healthcare delivery during public health crises, such as the COVID-19 pandemic?

- ☐ A. By eliminating the use of electronic health records (EHRs)
- ☐ B. By increasing the number of in-person appointments
- ☐ C. By making it more difficult for patients to receive prescriptions
- ☒ D. By reducing the need for in-person visits, thereby limiting exposure to infectious diseases

15. In the context of nursing informatics, "data integrity" refers to:
- A. Eliminating patient access to data
 - ☒ B. Ensuring that data is accurate, complete, and reliable for clinical use
 - C. Only gathering data from one specific source
 - D. Reducing the amount of patient data collected
16. What is the purpose of including a "knowledge management" component in the nursing informatics framework?
- A. To reduce the amount of information available to clinicians
 - B. To replace clinical judgment with automated decision-making
 - C. To restrict data use in non-clinical areas
 - ☒ D. To support the collection, organization, and sharing of knowledge for improving healthcare delivery
17. The Foundation of Knowledge Model supports clinical decision-making by:
- A. Limiting the amount of information available to healthcare providers
 - ☒ B. Organizing knowledge to be used effectively in practice
 - C. Preventing the integration of patient data
 - D. Replacing clinical judgment with automated systems
18. Which of the following describes "knowledge dissemination" in the Foundation of Knowledge Model?
- A. Accumulating information without sharing
 - B. Limiting knowledge access to specific professionals
 - ☒ C. Sharing and distributing knowledge to enhance understanding and practice
 - D. Storing knowledge in one location only
19. The goal of "knowledge processing" within the Foundation of Knowledge Model is to:
- A. Focus only on storing data without analysis
 - ~~B. Ignore irrelevant data~~
 - C. Limit knowledge to administrative purposes
 - ☒ D. Transform information and data into meaningful knowledge for clinical decision-making
20. How does the Foundation of Knowledge Model contribute to evidence-based practice in nursing?
- A. By excluding patient data from clinical decision-making
 - B. By focusing only on intuition in care practices
 - ☒ C. By providing a framework for acquiring and applying research-based knowledge in patient care
 - D. By reducing reliance on technological solutions
21. As a knowledge worker, a nurse primarily uses knowledge to:
- A. Avoid involvement in clinical decision-making
 - ☒ B. Improve patient care and support evidence-based practice
 - C. Focus solely on administrative documentation
 - D. Minimize communication with other healthcare providers
22. Which of the following best describes the role of technology in supporting the nurse as a knowledge worker?
- ☒ A. It enables the collection, storage, and easy retrieval of data for decision-making
 - B. It limits access to patient information
 - C. It replaces the need for nurses to analyze data
 - D. It solely supports administrative tasks in healthcare

32. What is one challenge associated with telemedicine implementation?

- A. Complete replacement of traditional healthcare settings
- B. Increased cost of in-person consultations
- C. Limited patient access to high-speed internet and necessary technology
- ☒ D. Reduced need for medical insurance

33. Telemedicine can improve healthcare outcomes by:

- A. Limiting the number of healthcare providers involved in a patient's care
- B. Reducing the amount of patient data collected
- ☒ C. Reducing the time it takes for patients to access medical care
- D. Replacing all in-person healthcare visits

34. Which of the following refers to the process of transforming raw data into meaningful insights or actionable information?

- A. Data mining
- B. Data compression
- C. Data encryption
- ☒ D. Data transmission

34. Which of the following is the definition of 'firewall' in information technology?

- A. A form of encryption for protecting health data
- B. A protocol for sharing health information between hospitals
- ☒ C. A software or hardware system designed to block unauthorized access to a computer network
- D. A type of storage device for patient data

Use the scenario below to answer question 36 to 40

A hospital has recently implemented a new Electronic Health Record (EHR) system. Nurses, doctors, and administrative staff use the system to document patient information, check test results, and manage appointments. After a month of using the system, staff members report that while it is generally functional, they find the system slow to load, especially when pulling up patient histories. Additionally, the system's alerts are sometimes too generic and hard to interpret, causing some confusion among healthcare providers. Some users have also complained that they need more training to become fully comfortable with the system.

36. What is the primary usability issue described in the scenario regarding the EHR system's slow loading times?

- A. Inadequate security measures for patient data
- B. Poor user interface design
- ☒ C. Poor system performance, specifically speed and response time
- D. Lack of integration with other systems

37. The system's alerts are described as too generic and hard to interpret. Which usability principle should be prioritized to resolve this issue?

- A. Enhancing the system's speed and response time
- B. Improving the clarity and specificity of alerts
- C. Increasing the system's encryption standards
- ☒ D. Making the system available in multiple languages

38. If the healthcare providers are struggling with the system due to a lack of training, which usability feature would best address this issue?

- A. Adding more advanced features to improve functionality
- B. Enhancing system encryption to increase security
- ☒ C. Providing in-depth, on-demand tutorials and training materials
- D. Restricting access to certain features for less experienced users

39. What usability issue is most likely causing confusion for healthcare providers when interpreting alerts?

- A. Inconsistent use of colors, icons, and text in alerts
- B. Lack of proper data encryption
- ☒ C. Lack of system feedback when actions are performed
- D. Overcomplicated interface design

40. Based on the scenario, which of the following actions would most likely improve the overall usability of the EHR system for the staff?

- A. Implementing a patient portal for staff communication
- B. Improving the system's data security features
- C. Increasing the frequency of system maintenance
- ☒ D. Simplifying the interface design and reducing the number of steps to complete tasks

SECTION B

1. List FIVE (5) principles of a good Patient Records Keeping

- * Use of EHR System
- * Provide Personal Record Keeping for Patient.
- * Clearly making rules for who can access Patient Information.
- * Telemedicine.
- * Evidence based practice.

2. Patients have specific rights under the Data Protection Act, including. List FIVE (5) of the Act.

- * Confidentiality.
- * Right to refusal of treatment.
- * Right for proper care treatment because they have paid for.
- * Right to access their information.
- * Right to be informed before any procedure is performed.